

HyPIM: Accelerating Hyperbolic Machine Learning via Processing-In-Memory

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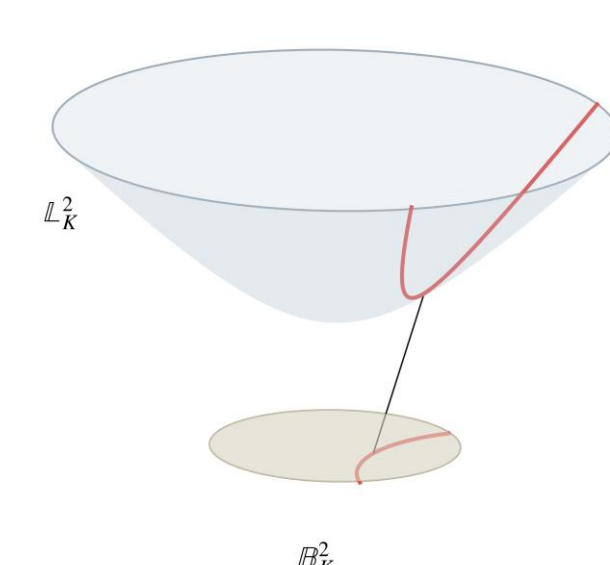
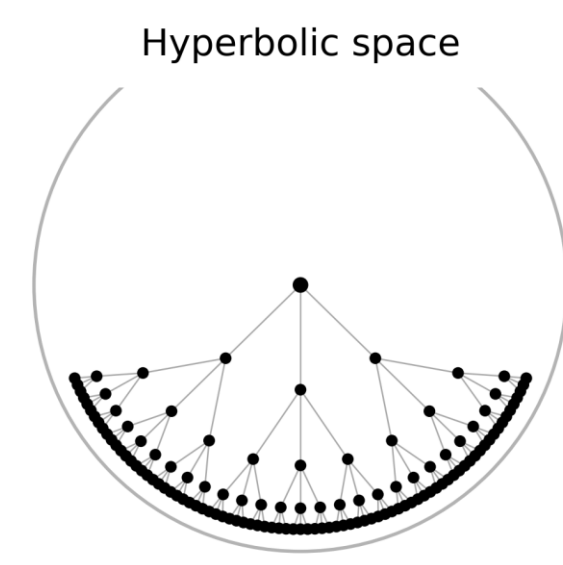
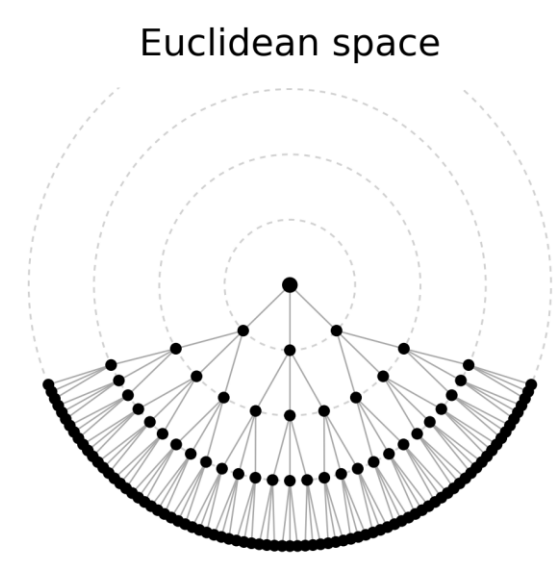
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Hyperbolic Space Machine Learning



Comparison of Tree Structure in Two Spaces

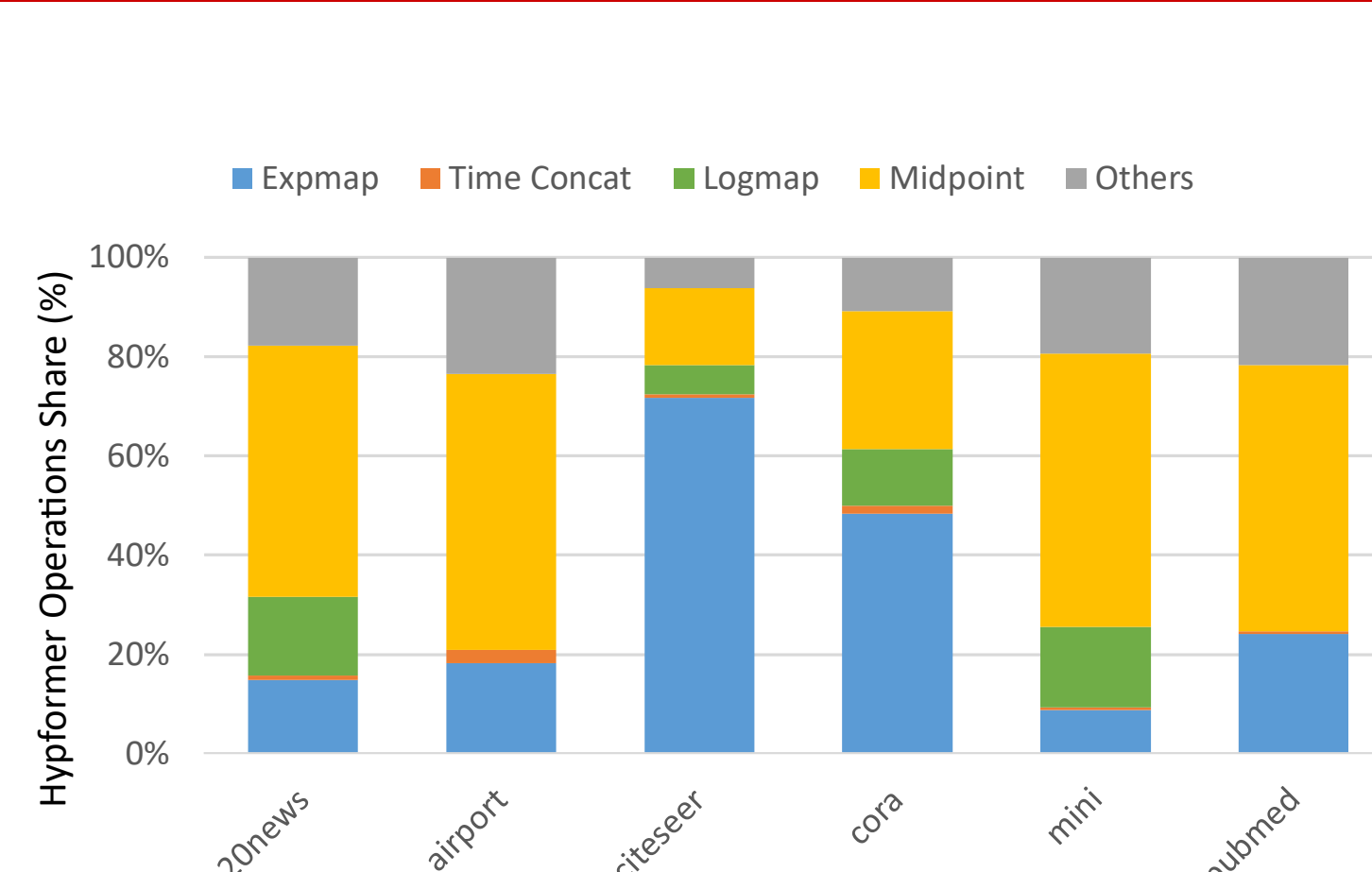
Project to Hyperbolic Space

Hyperbolic geometry provides advantages in machine learning over Euclidean space:

- Hyperbolic space has **exponentially expanding** volume.
- It naturally matches the exponential growth of tree-like structures, allowing nodes at deeper hierarchy levels to be represented with **lower distortion** introduced by crowding.
- Hyperbolic embeddings can preserve parent-child distances, sibling separation, and multi-level hierarchy structure with **fewer dimensions**.

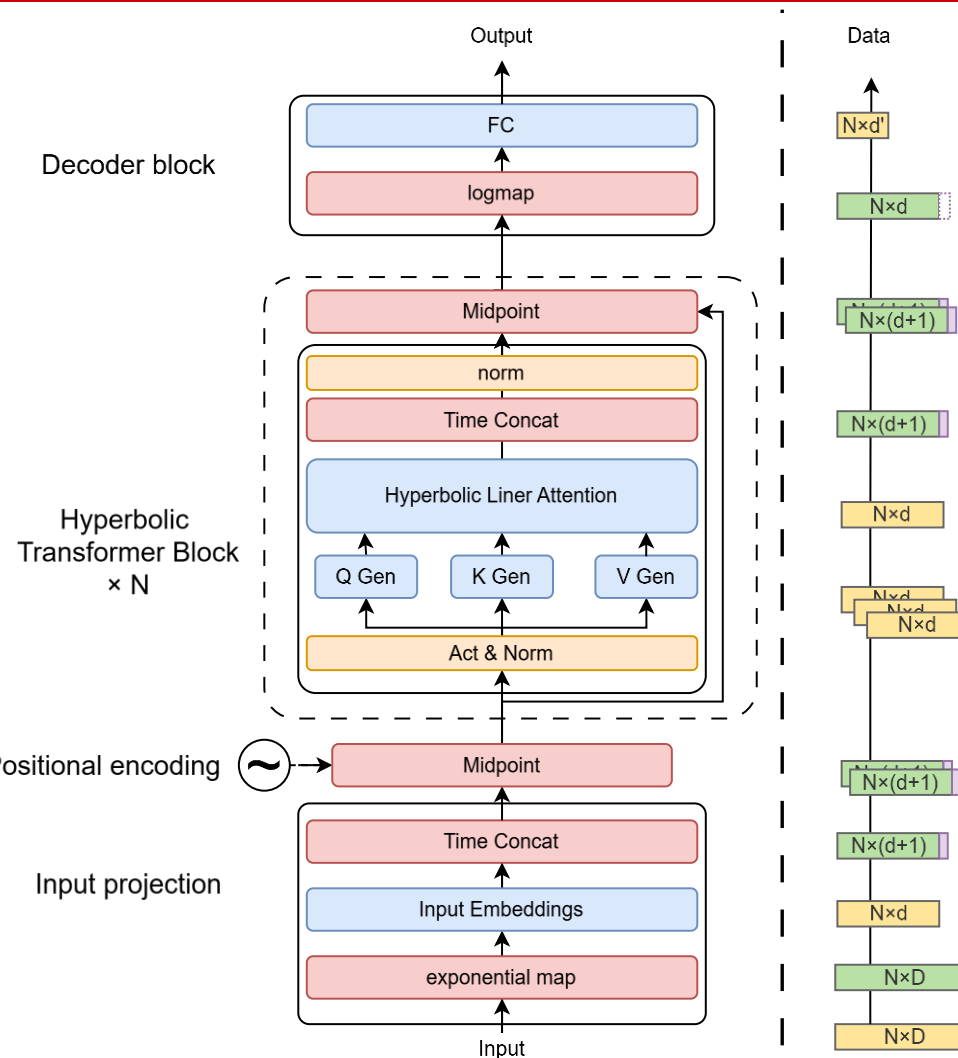
Researches like Hypformer (KDD'24) and HCNN (ICLR'24) have explored hyperbolic geometry as a new geometric foundation for machine learning and shown remarkable performance.

Challenge of Accelerating Hyperbolic Models



Operations Running Latency Share on GPU

Unlike matrix-multiplication kernels dominated conventional machine learning workloads, Hyperbolic ML models exhibit low arithmetic intensity and frequent reductions, leads to **80% GPU latency** spent on these operations.



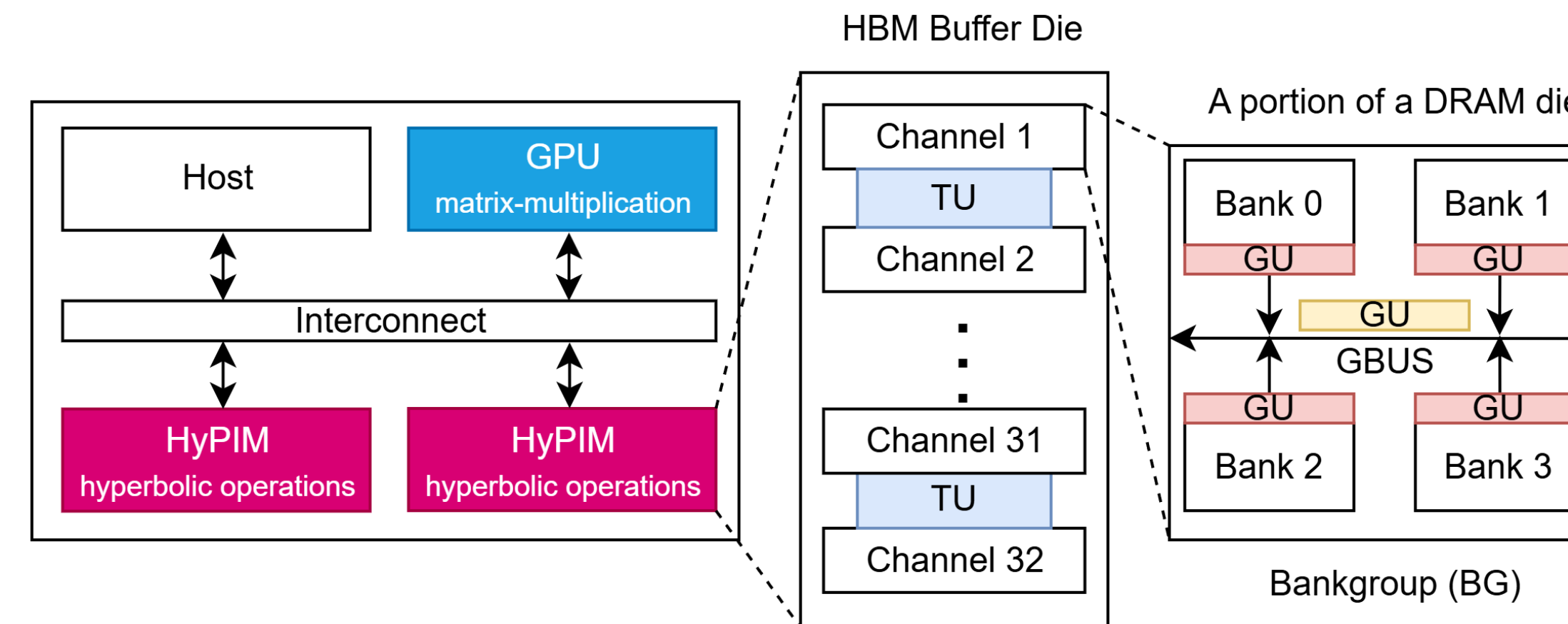
Hyperformer Model

Operation	Lorentz Model
Time concat	$x_t = \ x\ _2$
Exponential map (at the origin)	$\frac{1}{\sqrt{-k}} [\cosh(\sqrt{-k} x), \sinh(\sqrt{-k} x), \frac{x}{ x }]$
Logarithmic map (at the origin)	$\frac{1}{\sqrt{-k}} \text{arccosh}(x_t \sqrt{-k}) \frac{x}{ x }$
Midpoint (two points)	$\frac{\sqrt{2(1-k(-x_t y_t + x \cdot y))}}{\sqrt{-k} x _2} (x_t + y_t, x + y)$
Parallel transport (at the origin)	$\frac{x \cdot y}{\sqrt{-k} x _2} \frac{x}{ x _2} + \frac{x \cdot y}{ x _2^2} x$

Hyperbolic Operators

These hyperbolic ML models follow a common computational pattern: interleaving standard Euclidean layers with geometry-preserving transformations defined on the Lorentz model, requiring the hardware system to deal with both **compute-bound matrix-multiplication** and **memory-bound hyperbolic operations**.

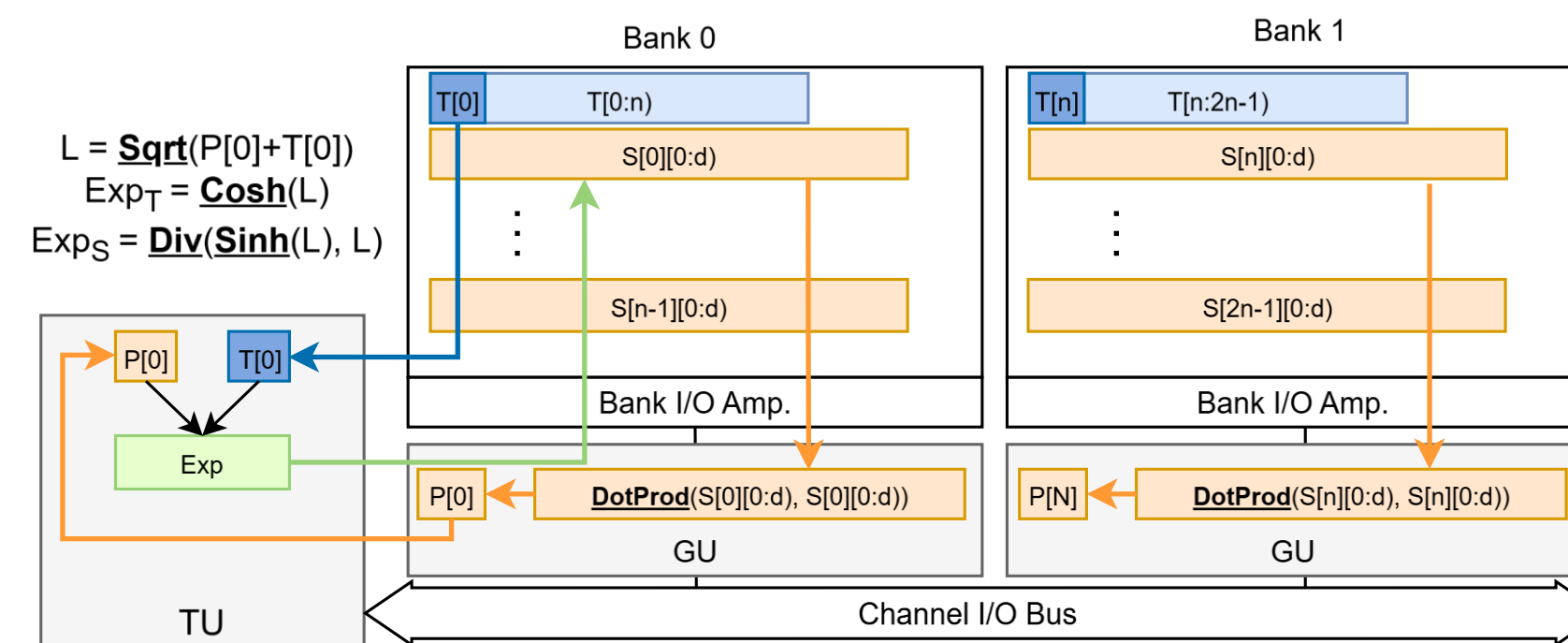
HyPIM Architecture



GPU-PIM Heterogeneous System PIM Heterogeneous System Offloading Hyperbolic Operations to PIM

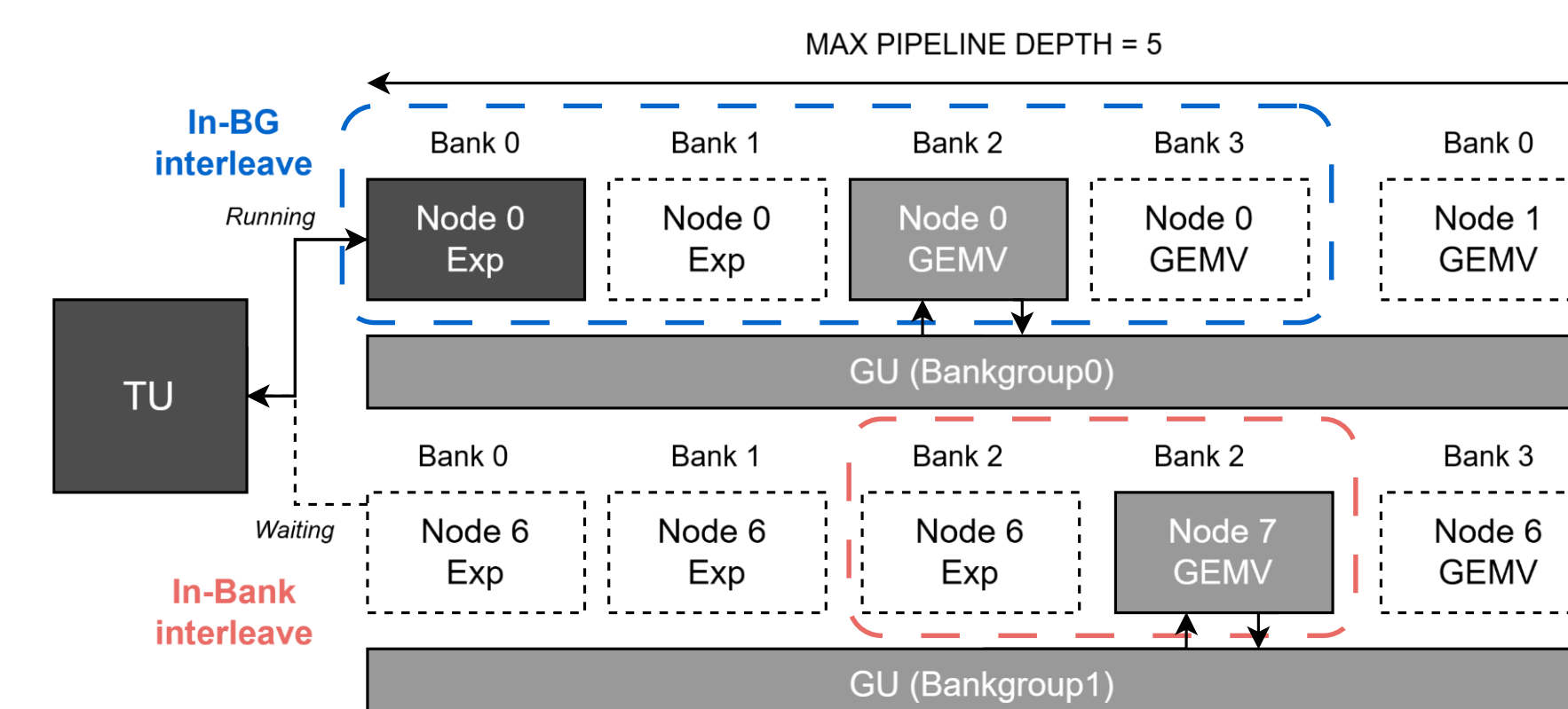
- Compute-dense operators remain on the GPU, while memory-bound hyperbolic operations execute inside HBM based PIM.
- GEMV Units (GUs) locate at each bank or bank-group level for elementwise multiplications and local reductions. The near bank placement avoid massive data movement in
- Transcendental Units (TUs) process scalar operators like exp and div at channel level, reaching a balance between operator density and hardware overhead for non-GEMV operators.

Data Layout and Scheduling Optimization



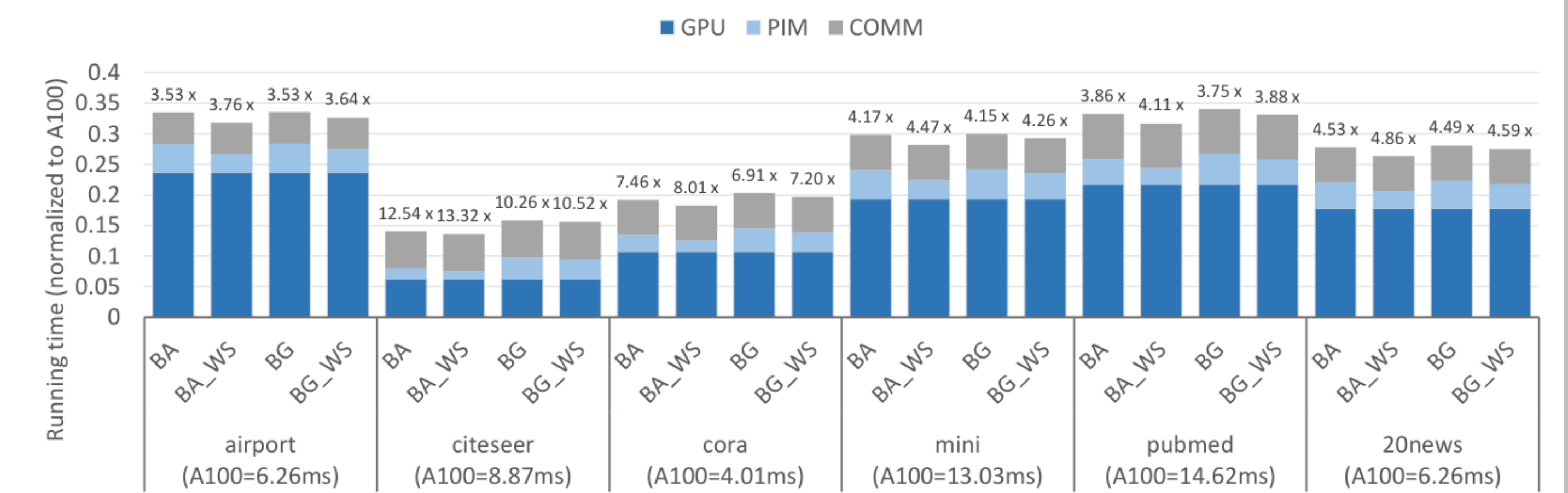
Expmap Execution Flow and Data Layout in PIM

The whole operation is split into stages and the layout enables fast TU access via the channel IO bus without interfering with GU vector streaming. In-flight scheduling will take advantages of them by issuing nodes of different stages to GUs and TUs to fully utilize hardware.

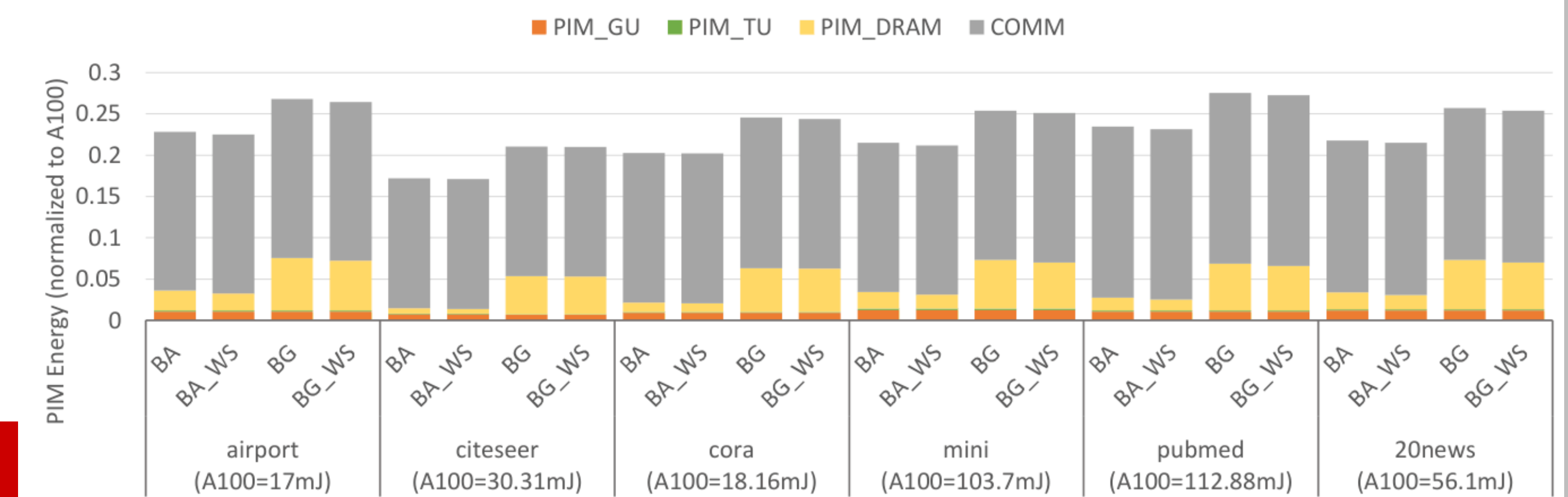


In-flight scheduling Issue different nodes to GU and TU

Evaluation



End-to-end Running Time Compared to A100 GPU Under 4 Configurations: Bank (BA)/Bank-group (BG) Unit Placement × Wavefront (WS)/None Scheduling



Energy Consumption Compared to A100 GPU

- On HypFormer datasets, HyPIM achieves 3~7× speedup an average of 30% total energy reduction in a single inference task.
- HyPIM leverages the high internal bandwidth and in-situ compute capability of HBM-PIM, allowing long feature vectors to be processed entirely within the memory stack without repeated off-chip transfers. As a result, workloads with large input dimensionality or broad per-node computation—such as Citeseer—benefit more substantially.

Conclusions and Impact

- We identify a fundamental compute–memory imbalance feature in Lorentz-model hyperbolic ML models.
- HyPIM, a GPU–PIM Heterogeneous System, and the first acceleration proposal for hyperbolic models.
- An optimized hyperbolic-friendly data layout and an in-flight wave-front scheduler.
- Evaluation on state-of-the-art hyperbolic neural networks for vision and graph datasets using ramulator2.